

Aditya Kumar

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EDUCATION

Dayananda Sagar College of Engineering

Bachelor of Engineering in Computer Science

Bengaluru, IN

Sep. 2025 – Sep. 2029

Kendriya Vidyalaya, Air Force Station

Chennai, IN

12th Standard — Percentage: 87%

2024

Kendriya Vidyalaya, Air Force Station

Chennai, IN

10th Standard — Percentage: 88.2%

2022

SKILLS

Languages: Golang ,Python ,JavaScript ,TypeScript ,HTML/CSS ,TailwindCSS

Database: PostgreSQL ,MongoDB

Frameworks: Next.js ,React ,Node.js ,Express

Developer Tools: Docker ,Postman ,Git ,Github ,Linux ,VS Code ,Vite

Libraries : WebRTC, Socket.IO ,React-map-gl ,spf13/cobra

PROJECTS

Dropa | *WebRTC, WebSockets, Next.js*

GitHub

- Built a P2P file transfer app using WebRTC DataChannels, which enables direct device-to-device file transfer with no upload/download size limits.
- Implemented chunked file transfer over encrypted RTCDataChannel with backpressure handling via bufferedAmountLow events, preventing memory overflows when transferring multi GB files across unreliable connections.
- Built a Socket.io signaling server to orchestrate SDP offer/answer exchange and ICE candidate trickle, enabling peers to establish connection.
- Improved user experience with dynamic QR codes tied to unique room IDs, reducing session setup to a single scan.

Armoracrypt | *Golang, Cobra, Dropbox-API*

GitHub

- ArmoraCrypt is a Golang-based CLI tool to facilitate local AES-256 encryption/decryption of files and folders, enabling secure uploads/downloads to cloud storage while safeguarding against provider surveillance and unauthorized access.
- Integrated zipping and unzipping functionality to handle entire folders, streamlining the encryption process for complex directory structures.
- Designed a master-key initialization command for generating and managing local encryption keys, ensuring a secure and self-contained key management system.
- Implemented upload and download features for encrypted files to cloud storage, allowing users to safely store and retrieve sensitive data.

Litspots | *Next.js, TailwindCSS, React-map-gl*

GitHub

- Built a full-stack real-time social discovery web app enabling users to find, join, and host live hangouts nearby.
- Reduced database load by caching recent API responses and serving stale data to prevent excessive queries.
- Integrated interactive maps with MapLibre GL and react-map-gl to display live hangout locations.
- Used a geocoding API to convert place names into geographic coordinates for accurate map-based discovery.
- Implemented backend APIs and custom authentication using Next.js API Routes, MongoDB, JWT, and bcryptjs.

Multiplayer Cursor | *Express, React, TailwindCSS, Vite, Socket.IO*

GitHub

- Developed a full-stack real-time mouse-sharing web application where users can see each other's cursor activities.
- Implemented both client and server logic using React and Express with Socket.IO for instant bi-directional communication.
- Added cursor smoothing to reduce jitter caused by network lag for a smoother experience.
- Used TailwindCSS to build a clean and lightweight interface.